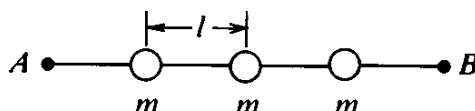


PHY201: Waves & Optics

Problem Set 5

September 20, 2025

1. An elastic string of negligible mass, stretched so as to have a tension T , is attached to fixed points A and B , a distance $4l$ apart, and carries three equally spaced particles of mass m , as shown.



- (a) Suppose that the particles have small transverse displacements y_1 , y_2 and y_3 respectively, at some instant. Write down the differential equation of motion for each mass.
 - (b) The appearance of normal modes can be found by drawing the sine curves that pass through A and B . Sketch such curves so as to find the relative values and signs of A_1 , A_2 , and A_3 in each of the possible modes of the system
 - (c) What is the time interval between successive maximum possible amplitudes of one pendulum after one pendulum is drawn aside and released?
2. Consider a system of N coupled oscillators driven at a frequency $\omega < 2\omega_0$ (i.e., $y_0 = 0$, $y_{N+1} = h \cos \omega t$). Find the resulting amplitudes of the N oscillators. [Hint: The differential equations of motion are the same as in the undriven case (only the boundary conditions are different). Hence try $A_p = C \sin \alpha p$, and determine the necessary values of α and C . (Note: If $\omega > 2\omega_0$, α is complex and the wave damps exponentially in space.)]
 3. A uniform rod is clamped at the center, leaving both ends free.
 - (a) What are the natural frequencies of the rod in longitudinal vibration?
 - (b) What is the wavelength of the n th mode?
 - (c) Where are the nodes for the n th mode?
 4. Derive the wave equation for vibrations of an air column. Your final result should be

$$\frac{\partial^2 \xi}{\partial x^2} = \frac{\rho}{K} \frac{\partial^2 \xi}{\partial t^2}$$

where ξ is the displacement from the equilibrium position, ρ is the mass density, and K is the elastic modulus. Show that for vibrations of an air column:

- (a) An open end represents a condition of zero pressure change during the oscillation, and hence a place of maximum movement of the air (an antinode).
 - (b) A closed end is a place of zero movement (a node) and hence maximum pressure variation.
5. (a) Find the total energy of vibration of a string of length L , fixed at both ends, oscillating in its n th characteristic mode with an amplitude A . The tension in the string is T and its total mass is M .

- (b) Calculate the total energy of vibration of the same string if it is vibrating in the following superposition of normal modes:

$$y(x, t) = A_1 \sin\left(\frac{\pi x}{L}\right) \cos \omega_1 t + A_3 \sin\left(\frac{3\pi x}{L}\right) \cos\left(\omega_3 t - \frac{\pi}{4}\right)$$

(You should be able to verify that it is the sum of the energies of the two modes taken separately.)

6. Find the Fourier series for the motion of a string of length L if

(a) $y(x, 0) = Ax(L - x)$; $(\partial y / \partial t)_{t=0} = 0$

(b) $y(x, 0) = 0$; $(\partial y / \partial t)_{t=0} = Bx(L - x)$.